

The V Index: A Multicriteria Metric for the Environmental Evaluation of LLM Inference

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Paper
ZENODO
zenodo.org/records/1973563



Code
COLAB
v3.0 batch notebook

1. Motivation

The era of massive inference. Since 2023, large language models have moved from research laboratories to billions of daily queries across search engines, productivity tools, and customer-service platforms.

Training is performed **once**; inference is executed **billions of times**. For deployed LLMs, the cumulative inference cost surpasses training within days to weeks of operation — a phenomenon we call the *Inference Gap*.

The Shift from Training to Massive Inference

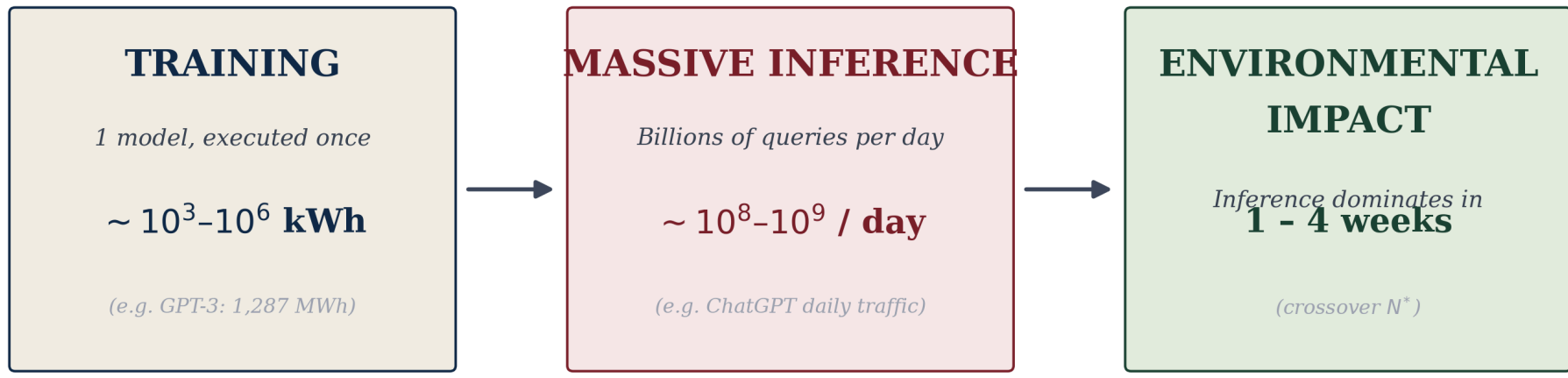


Fig. 1. Crossover point $N^* = C_{\text{train}} / \epsilon_{\text{inf}} t_{\text{h}}$ reached in days-weeks for all evaluated LLMs.

2. The Inference Gap

Existing metrics — **FLOPs**, **latency**, and **parameter count** — are indirect proxies. None captures real runtime energy, hardware efficiency, or the multicriteria environmental impact of a joule consumed in a given grid mix.

Carbon footprint (kg CO₂eq) is only one of **16 EF 3.1 impact categories**. The same inference in France (nuclear) and Singapore (natural gas) shows a **69.5×** difference in absolute environmental impact.

The Inference Gap: Why Standard Metrics Fail

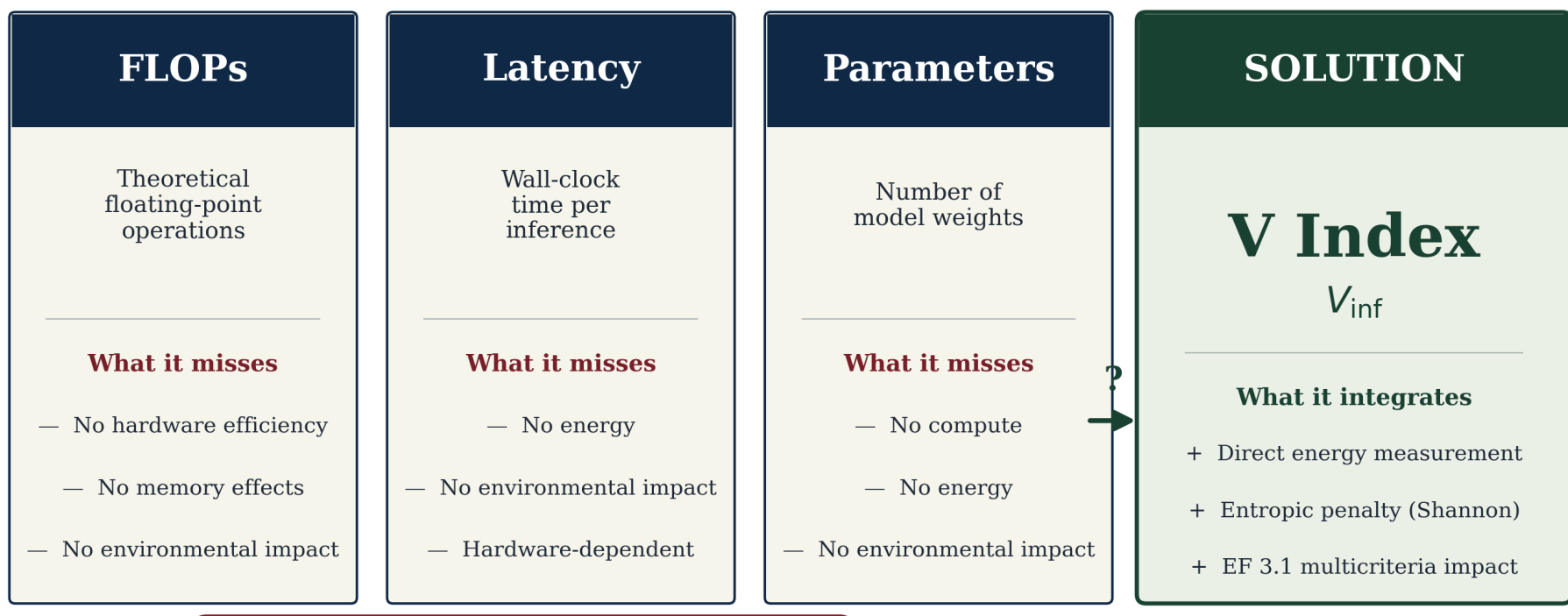


Fig. 2. Standard metrics fail to measure environmental sustainability. The V Index integrates direct measurement, entropy, and multicriteria impact.

3. Methodology

The pipeline separates autoregressive inference into two phases (**prefill** + **decode**), measures each via **NVML** on the GPU, computes per-token Shannon entropy to penalize redundancy, and aggregates through the EF 3.1 multicriteria factor γ .

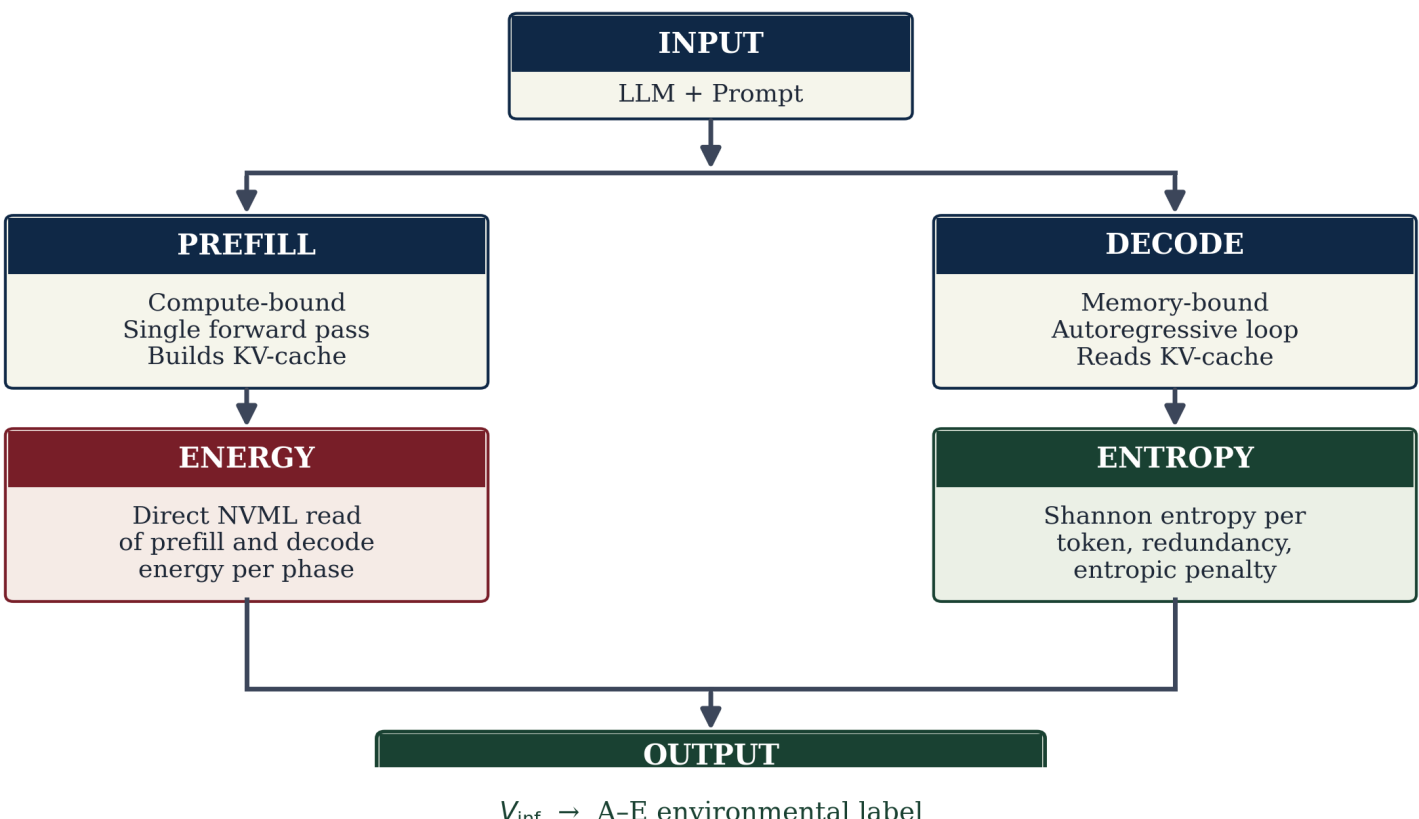


Fig. 3. End-to-end pipeline. Each phase is timed with `torch.cuda.synchronize()` for accurate GPU-energy readings.

4. Entropic Penalty Φ_{dec}

Each generated token has Shannon entropy $H_t = -\sum p \log_2 p$. The **redundancy rate** $R = 1 - H / H_{\text{max}}$ (with $H_{\text{max}} \approx 15.6$ bits for GPT-2) captures *babbling*. The penalty $\Phi_{\text{dec}} = 1 + \alpha \cdot R \cdot T_{\text{gen}} / T_{\text{ref}}$ ($\alpha = 0.7$) only penalizes, never rewards ($\Phi_{\text{dec}} \geq 1$).

Entropic Components by Quantization ($\Delta\Phi < 0.5\%$)

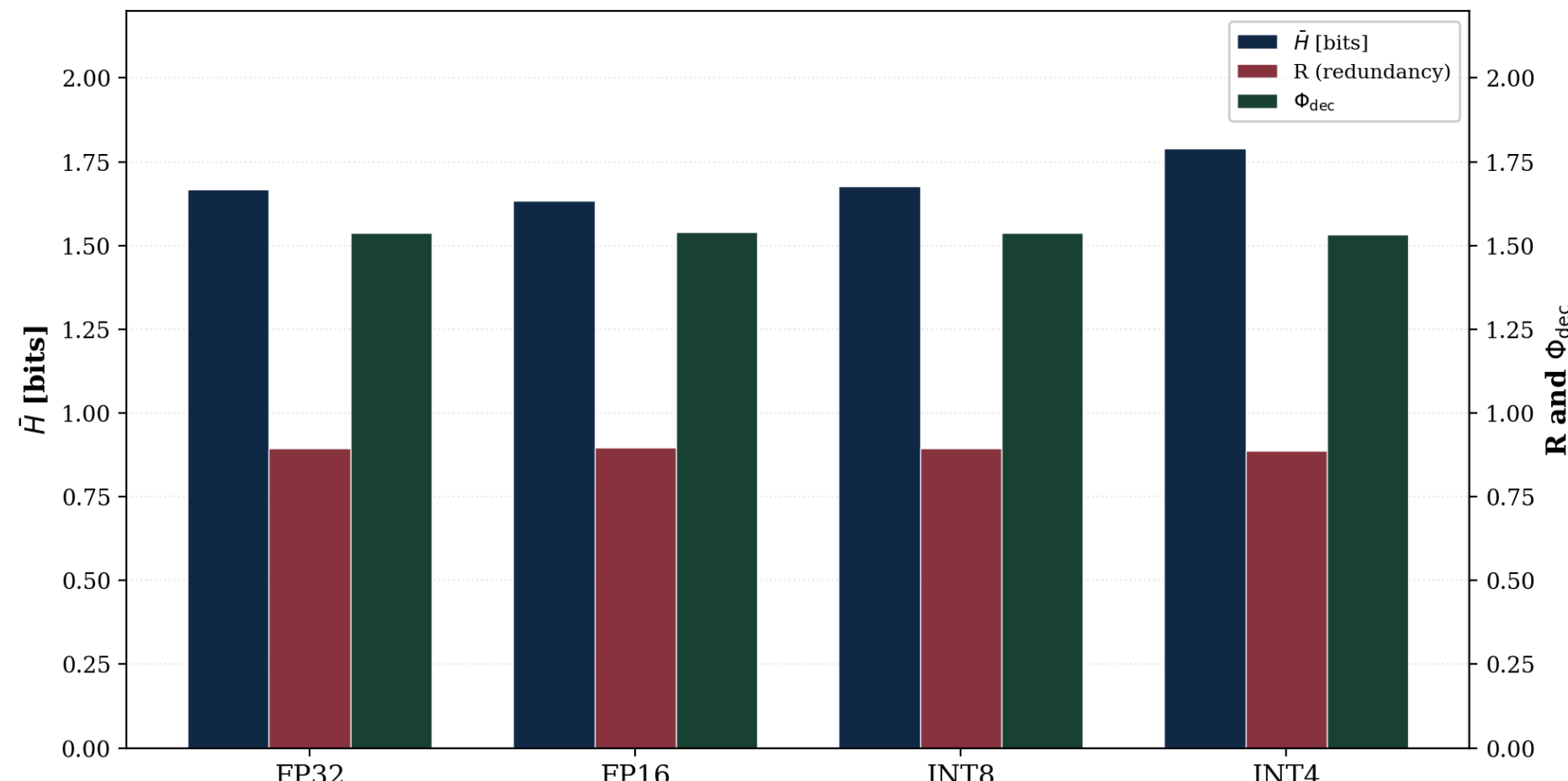


Fig. 4. Across quantizations, Φ_{dec} varies by less than 0.5% ($\in [1.532, 1.539]$) — the index discriminates configurations through the energy ratio, not the penalty.

5. Prefill vs. Decode Decomposition

The additive structure $V_{\text{inf}} = v_{\text{pre}} + v_{\text{dec}}$ reveals **which phase dominates** each configuration. In **93% of cases, decode dominates** ($> 60\%$ of V_{inf}), making decode-optimization techniques (batch decoding, speculative decoding, KV-cache compression) the highest-leverage interventions.

Additive Decomposition: Prefill + Decode

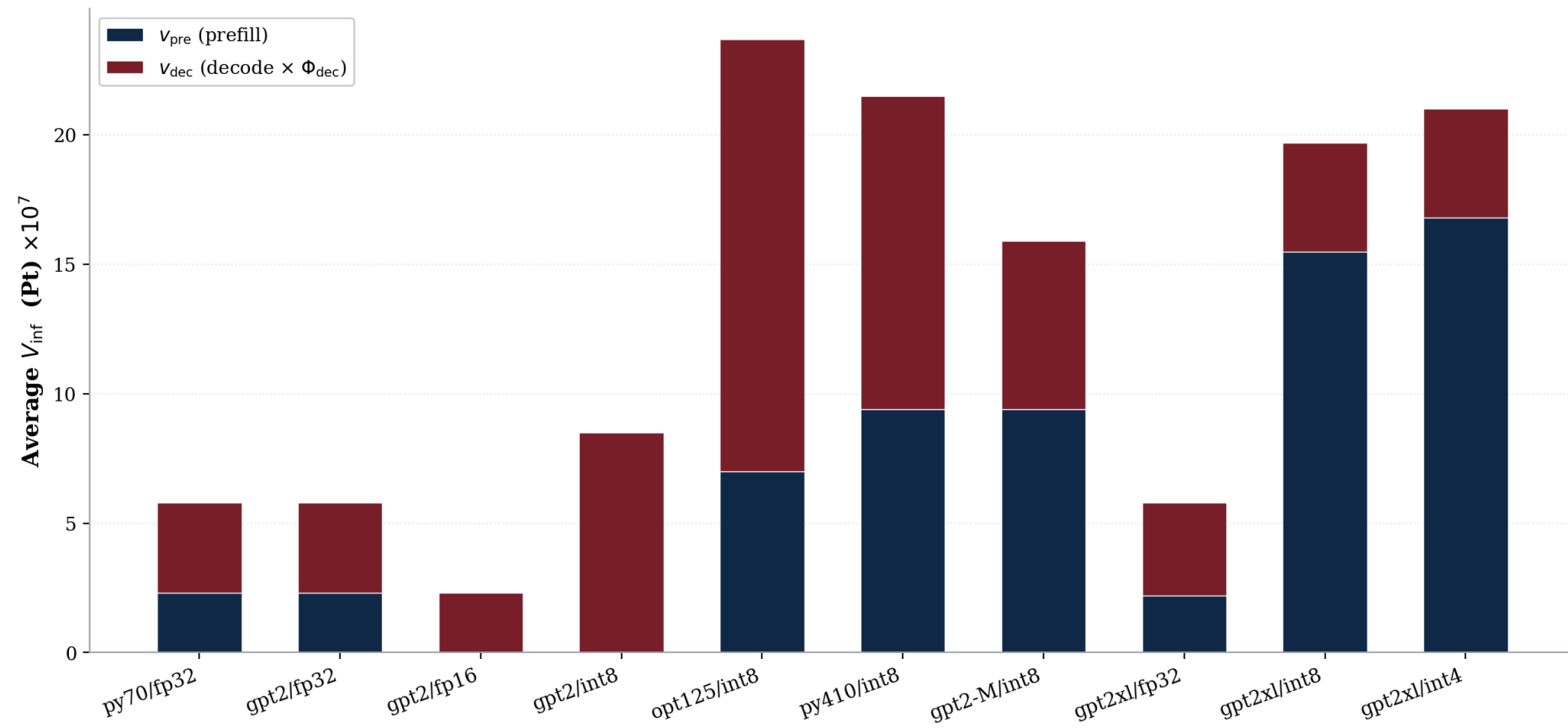


Fig. 5. Decomposition of V_{inf} into prefill (blue) and decode (red, including Φ_{dec}). Large quantized models (gpt2-3.1/int8, gpt2-3.1/int4) are the exception: prefill contributes $> 78\%$ due to the high $E_{\text{pre}}/E_{\text{dec}}$ ratio of the 1.5B-parameter model.

6. Why This Matters

The V Index translates a question that *FLOPs cannot answer*: **is this configuration environmentally responsible?** By separating the dominant cost (decode) from the entry cost (prefill), it tells practitioners where to invest optimization effort.

Practitioner guidance:

- Label [A]:** ship to production; baseline validated.
- Label [B]:** acceptable; monitor at scale.
- Label [C]:** review quantization strategy; consider FP16.
- Label [D]:** redesign before scaling; energy grows faster than throughput.
- Label [E]:** blocking; configuration is not viable for production.

★ The V Index ★

$$V_{\text{inf}} = \sum_{h \in \{\text{CPU, GPU}\}} \left(\frac{E_{\text{pre}}^h}{E_{\text{pre}}^* h} \gamma_h + \frac{E_{\text{dec}}^h}{E_{\text{dec}}^* h} \Phi_{\text{dec}}(\hat{Y}) \gamma_h \right)$$

Additive over phases & hardware · multiplication over energy × entropy × impact

$$V_{\text{inf}} \approx \gamma_{\text{GPU}} \left(\underbrace{\frac{E_{\text{pre}}^{\text{GPU}}}{E_{\text{pre}}^* \text{GPU}}}_{v_{\text{pre}}} + \underbrace{\frac{E_{\text{dec}}^{\text{GPU}}}{E_{\text{dec}}^* \text{GPU}}}_{v_{\text{dec}}} \Phi_{\text{dec}}(\hat{Y}) \right)$$

$$V_{\text{rel}} = \frac{V_{\text{inf}}}{V^*} = \frac{V_{\text{inf}}}{2 \gamma_{\text{GPU}}} \quad V_{\text{rel}} = 1 \iff \text{baseline}$$

Three components, one metric:

ENERGY Direct NVML measurement, separated by phase (prefill = compute-bound, decode = memory-bound) and by hardware.

ENTROPY Shannon-redundancy penalty. Only penalizes, never rewards. Calibrated with $\alpha = 0.7$.

IMPACT EF 3.1 multicriteria conversion to Points (Pt/J) over 5 categories (climate, fossil, water, particulates, eutrophication).

SYMBOL	NAME	DESCRIPTION	UNIT
$E_{h,p}$	Energy	Measured energy on hardware h , phase p	J
$E_{h,p}^*$	Baseline	Efficient reference (FP32 of each model)	J
Φ_{dec}	Entropic penalty	Penalizes redundant generation	≥ 1
γ_b	Env. factor	Joules $\rightarrow$ EF 3.1 Points	Pt/J
V_{rel}	Relative V	Normalized index (1.0 = baseline)	—

7. Experimental Setup

Experimental Setup

MODELS	QUANT.	HARDWARE	LOCATIONS	EVALUATIONS
12	4	T4	7	315
GPT-2, OPT, Pythia, BLOOM, DistilGPT-2, Qwen2	FP32, FP16, INT8 (LLM.int8()), INT4 (NF4)	NVIDIA T4, 16 GB VRAM, 70 W TDP, NVML level 1	France, Oregon, Germany, Virginia, Texas, Argentina, Singapore	45 configurations × 7 geographic locations
$T_{\text{gen}} = 128$ tokens · $N = 3$ benchmark iterations · Baseline: FP32 per model				

Fig. 5. 12 autoregressive models from 6 architectural families (GPT-2, OPT, Pythia, BLOOM, DistilGPT-2, Qwen2); 4 quantization modes; 7 geographic locations with carbon intensities spanning two orders of magnitude; **315 total evaluations**. Baseline: FP32 of each model. $T_{\text{gen}} = 128$ tokens, $N = 3$ averaged iterations.

8. A–E Efficiency Labels

Inspired by **EU energy regulation** for household appliances, the V Index produces a directly interpretable label: **[A]** optimal $\rightarrow$ **[B]** $\rightarrow$ **[C]** $\rightarrow$ **[D]** $\rightarrow$ **[E]** critical.

A-E Environmental Efficiency Labels

[A]	[B]	[C]	[D]	[E]
Optimal	Efficient	Moderate	Inefficient	Critical
$V_{\text{rel}} < 1.0$	$1.0 \leq V_{\text{rel}} < 1.5$	$1.5 \leq V_{\text{rel}} < 2.5$	$2.5 \leq V_{\text{rel}} < 4.0$	$V_{\text{rel}} \geq 4.0$
13.3%	35.6%	26.7%	13.3%	11.1%

Inspired by EU energy regulation for household appliances — interpretable for AI deployment decisions

Fig. 6. Label distribution over 315 evaluations. The index discriminates across the full A–E scale [13.3% [A] · 35.6% [B] · 26.7% [C] · 13.3% [D] · 11.1% [E]].

8. Comparison with Existing Metrics

FEATURE	FLOPS	TOK/S	KG CO ₂	CODECARBON	V_{inf}
Direct energy measurement	no	no	no	partial	yes
Prefill / decode separation	no	no	no	no	yes
Redundancy penalty (entropy)	no	no	no	no	yes
Multicriteria impact (EF 3.1)	no	no	no	no	yes
Decomposable by hardware	no	no	no	partial	yes
Interpretable A–E label	no	no	no	no	yes
Geographic sensitivity	no	no	yes	yes	yes

Table 1. Feature comparison. The V Index is the only metric that integrates direct phase-level measurement, entropic penalty, and multicriteria impact.

9. The EF 3.1 Framework

The environmental factor γ converts joules to **EF 3.1 Points (Pt)** by aggregating five normalized impact categories with the JRC's weighting factors, scaled by the regional carbon intensity CI_{grid} and the datacenter PUE.

EF 3.1 Multicriteria Environmental Factor γ

Five impact categories aggregated into a single score (Points / Joule)

$$\gamma_h = k_{\text{full}} \cdot \sum_{i=1}^5 \frac{c_i^{\text{ref}}(h)}{W_i} \cdot W_i, \quad k_{\text{full}} \approx 2.05$$

#	Category	Formula	Unit	w
1	Climate change	$CI \times PUE$	kg CO ₂ -eq	0.2106
2	Fossil resources	$\rho \times PUE$	MJ	0.0832
3	Water use	$WUE \times PUE$	m ³	0.0851
4	Particulate matter	$CI \times PUE$	disease inc.	0.0896
5	Aquatic eutrophication	$CI \times PUE$	mol N eq	0.0281

CI_{grid} = regional carbon intensity · PUE = Power Usage Effectiveness · WUE = Water Usage Effectiveness · ρ = fossil fraction

Fig. 8. The five impact categories used in the V Index. Climate change dominates by weight (0.21); the projection factor $k_{\text{full}} \approx 2.05$ extrapolates the score to the full 16-category EF 3.1 framework.

10. Limitations & Caveats

- Cloud energy bounds.** RAPL is locked in Google Colab; CPU energy is estimated via $TDP \times \text{usage} \times \text{duration}$ (±30%). GPU uses NVML level 1 (±5%).
- NVML granularity.** 1 mJ resolution, 20 ms update window. Very short prefill phases (< 50 ms) can return 0.0 (handled via the v3.3 falsy-zero guard).
- Static emission factors.** Annual-average CI_{grid} real-time marginal data (Electricity Maps API) would refine precision.
- Simplified embodied component.** $M_{\text{CC}} \approx 80$ kg CO₂-eq for a T4 (literature estimate). Contributes < 0.01% of the total factor.
- Prompt and engine dependence.** Results vary with prompt and inference engine (vLLM, TensorRT-LLM, DeepSpeed, Transformers) — a future benchmark dimension.

12. Results — Efficiency Ranking

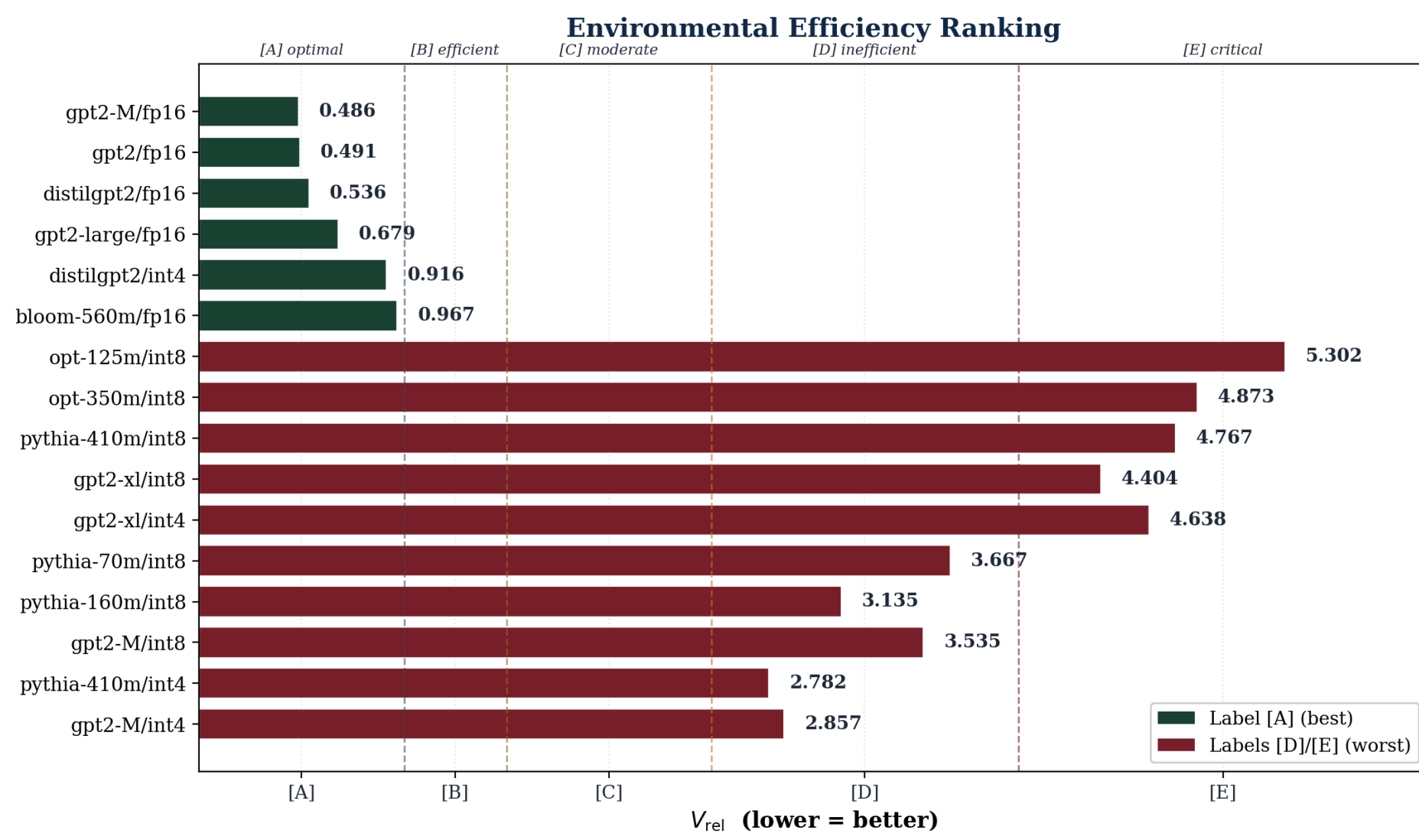


Fig. 7. Environmental efficiency ranking: 6 best (label [A]) and 10 worst (labels [D]/[E]) configurations. All 6 best are **FP16**; the 5 worst are all **INT8**. Best overall: **gpt2-medium/fp16** ($V_{\text{rel}} = 0.486$).

13. Results — Decode Energy

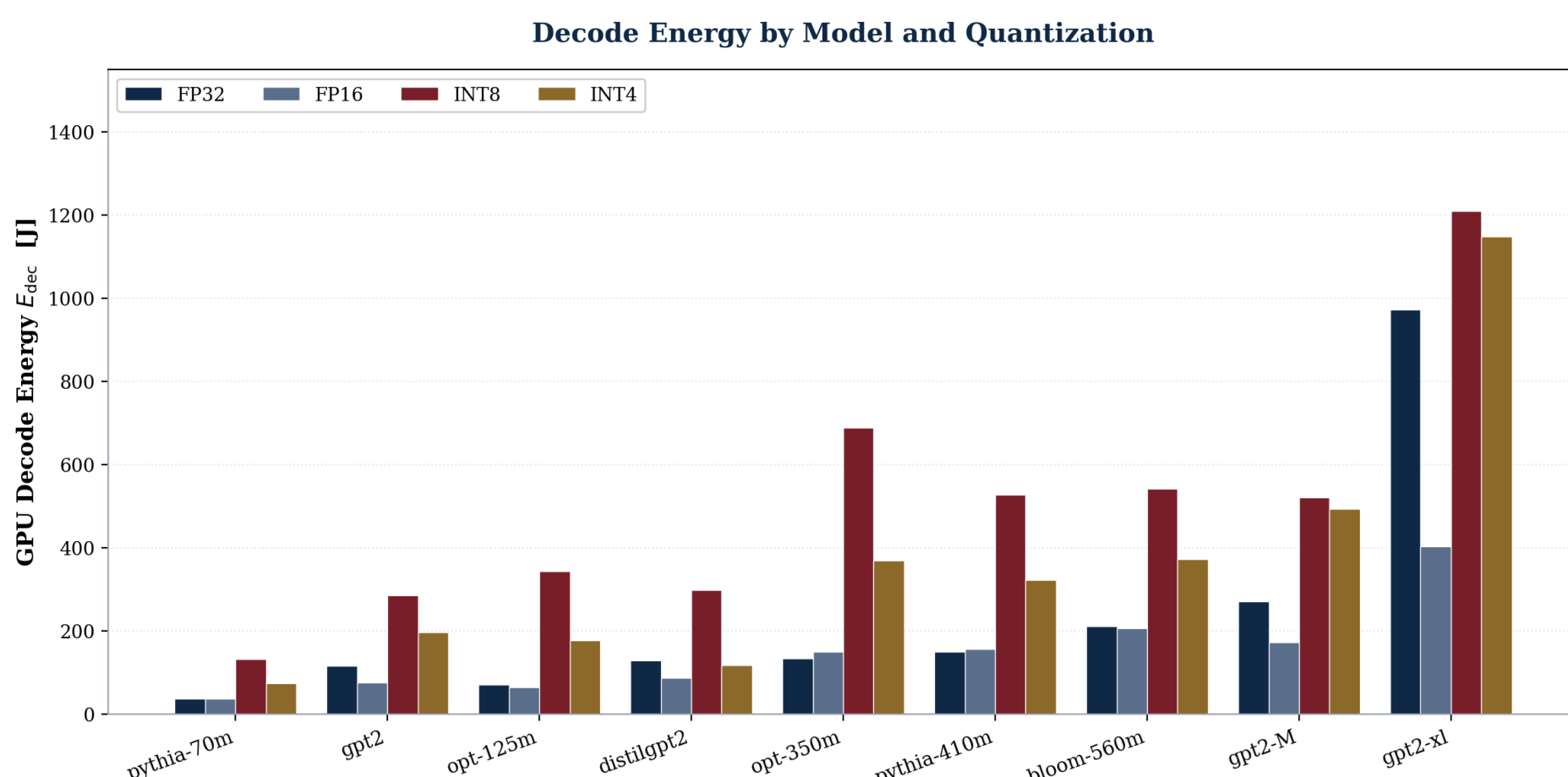


Fig. 8. CPU energy in the decode phase ($T_{\text{gen}} = 128$ tokens, NVML). **INT8 surpasses FP32** in every model. **FP16** is systematically the most efficient thanks to native Tensor Cores on T4. Average: FP16 = 163 J, FP32 = 249 J, INT4 = 389 J, INT8 = 515 J.

14. Results — Geographic Sensitivity

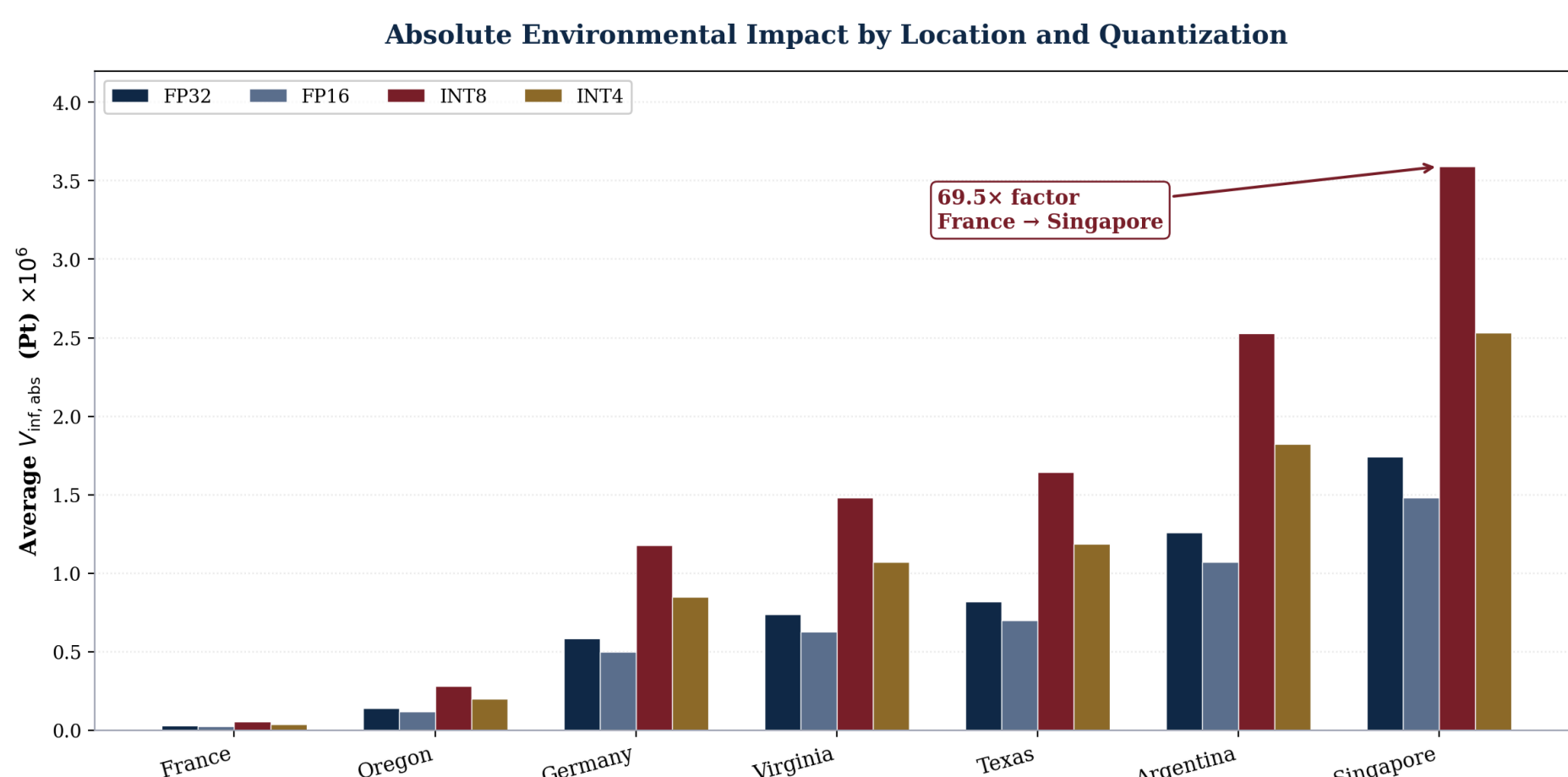


Fig. 9. Absolute $V_{\text{inf},\text{dec}}$ by location and quantization (12-model average, $P_t \times 10^6$). The **69.5×** factor between France (nuclear) and Singapore (natural gas) broadly exceeds the 10.9× factor between worst and best quantization. V_{rel} is **geographically invariant** ($\sigma = 0.0$).

★ Key Findings

- FP16 is best on T4.** $V_{\text{rel}}^{\text{FP8}} = 1.078$; all 6 [A] configurations are FP16. Best: gpt2-medium/fp16 ($V = 0.486$).
- INT8 is pathological on T4.** $V_{\text{rel}}^{\text{FP8}} = 3.485$ (2.75× worse than FP32). The 5 worst are all INT8 — invisible to FLOPs.
- Geography dominates.** Singapore/France factor = 69.5× — datacenter relocation is more effective than quantization optimization.
- Decode dominates (93%).** In 42/45 configurations, decode contributes more than 60% of V_{inf} .
- INT4 is model-dependent.** $\sigma = 0.926$: distilgpt2/int4 reaches [A] (0.916); gpt2-xl/int4 falls to [E] (4.638).
- Entropy is stable.** $\Phi_{\text{dec}} \in [1.532, 1.539]$ ($\Delta < 0.5\%$) — discrimination comes from the energy ratio, not the penalty.
- Full A–E discrimination.** After the v3.2 baseline correction, the index covers the complete scale across all 315 evaluations.

15. Conclusions

- First direct, multicriteria, decomposable metric** for the environmental cost of LLM inference — combining runtime measurement, information theory, and EF 3.1.
- Additive over phases and hardware;** multiplicative over energy × entropy × environmental factor.
- Validated** on 12 models, 4 quantizations, 7 locations, 315 evaluations.
- Provides a **formal, reproducible foundation** for sustainable AI deployment decisions.

16. Future Work

- Larger models** (1B–7B, LLaMA, Mistral, Phi-2) on A100/H100 with native INT8/FP8.
- Cross-family baselines** for inter-family comparison.
- Dynamic carbon intensity** via Electricity Maps API.
- Multimodal extension** (vision-language models).
- Energy–quality Pareto frontiers** across model families.
- Active babbling suppression** (Solyoyeva & Castor, 2026).

17. Key References

- Strubell, Ganesh & McCallum (2019). Energy and Policy Considerations for Deep Learning in NLP. *ACL*, 3645–3650.
- Luccioni, Viguer & Ligozat (2023). Estimating the Carbon Footprint of BLOOM. *JMLR*, 24(253), 1–15.
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- Sala et al. (2018). EF 3.1 Characterization Factors. *JRC Tech. Rep.*, European Commission.
- Fernandez et al. (2025). Energy Considerations of LLM Inference. *ACL*, 32556–32569.
- Solyoyeva & Castor (2026). Towards Green AI: Decoding LLM Inference Energy. *arXiv:2602.05712*.

18. Reproducibility

Hardware. All experiments on Google Colab with an NVIDIA T4 GPU (16 GB VRAM, 70 W TDP, Turing architecture, $M_{\text{CC}} = 80$ kg CO₂eq).

Protocol. $T_{\text{gen}} = 128$ tokens, $N = 3$ averaged iterations, standardized prompt, FP32 baseline per model. NVML level 1 with `torch.cuda.synchronize()` before each reading.

Quantization. BitsAndBytes (LLM.int8()) for INT8, NF4 with double quantization for INT4) applied per model; Qwen2.0.5B evaluated in FP16 only.

Data sources. AIB, EPA eGRID, ENTSO-E, CAMMESA, EMA, IEA — annual average CI_{grid} for each of the 7 geographic locations.

Code & data. Open implementation on GitHub (see QR code); extended results and per-configuration tables available in the paper (Table 15, Zenodo record 1973563).